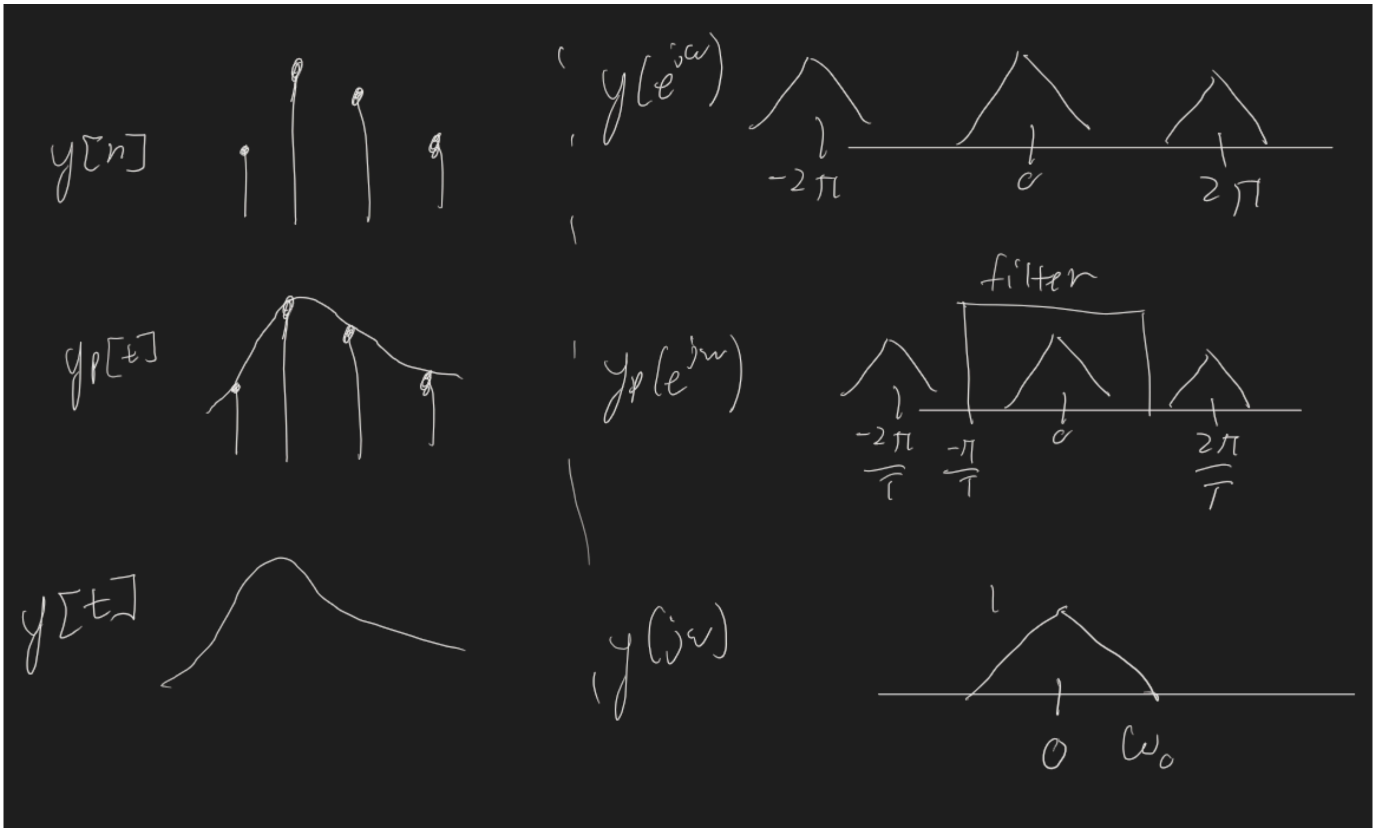
18 Sampling, Digital to continuous

Last time, we build a continuous to digital filter. Now we need to convert back, ideally an exact reconstruction of what we did before?



This works because we have the reconstruction filter, the thing that is a boxcar from $-\frac{\pi}{T}$ to $\frac{\pi}{T}$.

$$\boxed{\begin{aligned} h(t) &= \frac{1}{2\pi}\int_{-\infty}^{\infty} H(j\omega)e^{j\omega t}d\omega \\ &= \frac{T}{2\pi}\int_{-\frac{\pi}{T}}^{\frac{\pi}{T}} e^{j\omega t}d\omega \\ &= \frac{T}{2\pi}\frac{1}{jt}e^{j\omega t}\Big|_{-\frac{\pi}{T}}^{\frac{\pi}{T}} \\ &= \frac{T}{2\pi jt}\left(e^{\frac{jt\pi}{T}} - e^{-j\frac{t\pi}{T}}\right) \\ &= 2j\frac{\sin\left(\frac{\pi}{T}t\right)}{\frac{\pi t}{T}} \end{aligned}}$$

The practical realistic solution, (because this goes from -inf to inf) is to have a zero order hold, where at each sample instant we hold it and make the next sample. This has an impulse response $h(t)$ that looks like a box of width T from 0 to T with height 1.